Saliency Analysis and Gaussian Mixture Model-Based Detail Extraction Algorithm for Hyperspectral Pansharpening

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***Abstract*— The purpose of hyperspectral (HS) pansharpening is to improve the spatial resolution of HS images using panchromatic (PAN) images so as to obtain the pansharpened images with both high spectral diversity and high spatial resolution. The classical component substitution (CS)-based pansharpening approaches can be decomposed into two sequential phases: detail extraction and detail injection. In general, the detail extraction is performed by computing the difference between the PAN image and a weighted average of the HS bands, whereas the detail injection depends on the injection gain which is defined locally or globally. In this article, we introduce a novel pansharpening algorithm in which the extracted details and the injection gain are estimated over salient and nonsalient regions achieved via saliency analysis and Gaussian mixture model. The proposed method is applied to four credible CS-based pansharpening methods and also compared to other state-of-the-art methods. Experimental results show that the modified CS methods have better performance than the original methods. In addition, our method also achieves comparable or better performance than the other state-of-the-art methods.**

***Index Terms*— Detail extraction, Gaussian mixture model (GMM), pansharpening, saliency analysis.**

I. Introduction

REMOTE sensing image fusion combines remote sensing images under the same scene taken by different angles, different resolutions, or different sensors to obtain the fused higher quality images [1]. As a key preprocessing step for the application of remote sensing images, remote sensing image fusion has received attention in many fields such as geological survey, environmental monitoring, and urban planning [2]. Among the remote sensing image fusion techniques, hyperspectral (HS) pansharpening has become an active area of research in recent years. HS imagery with high spectral diversity can be used to distinguish the physical nature of different materials [3], but its low spatial resolution limits its applications severely. On the other hand, panchromatic (PAN) images with a high spatial resolution are often used to locate targets of interest. The goal of HS pansharpening is to improve the spatial resolution of HS images by using the spatial information of PAN images to obtain pansharpened images with both high spectral diversity and high spatial information that cannot be acquired by a single sensor due to the costs and physical limitations. The pansharpened HS images not only improve the accuracy of image processing, such as classification and detection, but also extend the range of HS image applications. With the increasing use of commercial products that use high-resolution images, such as Google Earth, the demand for pansharpened images is growing. Therefore, pansharpening has attracted more and more interest from the scientific community.

Many pansharpening methods have been proposed for addressing the pansharpening problem [4], [5]. Alparone *et al.* [6] reviewed the development of pansharpening over the past two decades. The HS pansharpening techniques are generally divided into "classical pansharpening methods" based on simple fusion models and recently proposed algorithms based on complex models. The Bayesian and matrix factorization approaches are the most recent algorithms and have the advantages of good spatial performance and high spectral fidelity. For the scene of interest, the Bayesian fusion model defines an appropriate prior distribution of the full resolution image to regularize the ill-posed fusion problem. As a special case of the Bayesian method, the matrix factorization [7] approach estimates the high-resolution HS images by maximizing the defined prior distribution. Bayesian Naïve Gaussian prior (Bayesian Naïve) [8], Bayesian sparsity promoted Gaussian prior (Bayesian sparse) [9], [10], Bayesian HySure [11], and the coupled nonnegative matrix factorization (CNMF) [12] are the most popular algorithms of Bayesian and matrix factorization approaches. However, the Bayesian and matrix factorization approaches all have the shortcoming of high complexity; hence they are not applicable for real-time applications.

The classical pansharpening algorithms can usually be divided into two consecutive steps of detail extraction and detail injection. Detail extraction is the extraction of spatial

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details for improving the spatial resolution of HS images. Detail injection is to inject the extracted spatial details into the interpolated HS image. The detail extraction is a method-dependent procedure, which depends on the way the detail image is estimated. According to the difference of detail extraction, the classical algorithms can be classified into component substitution (CS)-based algorithms and multiresolution analysis (MRA)-based algorithms. Among them, the CS-based algorithms extract the difference between the PAN image and a weighted average of the HS bands. Some popular methods belonging to this family are intensity-hue-saturation (IHS) [13], generalized intensity-hue-saturation (GIHS) [14], Brovey transform (Brovey) [15], Gram–Schmidt (GS) [16], adaptive GS (GSA) [17], and principal component analysis (PCA) [18], [19]. The MRA-based algorithms extract the difference between the PAN image and its low-frequency estimation. Relevant works of this class are the modulation transfer function-generalized Laplacian pyramid (MTF-GLP) [20], MTF-GLP with high pass modulation (MTF-GLP-HPM) [21], and smoothing filter-based intensity modulation (SFIM) [22] methods.

According to the definition of the injection gain, the detail injection can be divided into two classes: 1) global injection methods, which use a single injection coefficient for each band of an image and 2) context-adaptive methods, which exploit spatially variant injection coefficients for the whole image. The global injection methods are simple and their computational complexity is low, whereas the context-adaptive methods can obtain better performance by locally deriving the injection coefficients [23]. To make a tradeoff between computational complexity and performance, Wang *et al.* [24] introduced a nonoverlapping windows-based local method. Restaino *et al.* [25] proposed an image segmentation-based local coefficients estimation method, which has been integrated into GS orthogonalization procedure-based CS class and the GLP-based MRA family.

The basic principle of CS-based pansharpening algorithms is to replace the spatial component of HS image with PAN image. They were originally proposed for the fusion of PAN and multispectral (MS) images and are popular for their simplicity and efficiency. However, the MS and PAN images almost cover the same spectral range, while HS images are usually acquired in a much wider spectral range. Therefore, the CS-based HS pansharpening methods usually suffer more severe spectral distortion than the MS pansharpening algorithms due to spectral range mismatch between HS and PAN images. Moreover, another drawback is that the low-pass component of the PAN image may also cause spectral distortion of the pansharpened image. In this article, we propose a new region pansharpening method, which aims at reducing the spectral distortion caused by the low-pass component of the PAN image and the spectral mismatch between the HS image and the PAN image. The advantages of the proposed algorithm are highlighted as follows.

1) Traditional CS-based pansharpening algorithms directly match and replace the spatial component of the HS image with the PAN image. Thomas *et al.* [26] proposed that the greater the correlation between the replaced component and the PAN image, the smaller the spectral distortion caused by this pansharpening method. Inspired by this result, we propose to fuse the details of PAN and that of HS images to get the total detail map and use this total detail map to match and replace the spatial component of HS image. The total detail map considers the structures of both PAN and HS images and is thus more suitable for substituting the spatial component of the HS image in pansharpening.
2) To reduce the spectral distortion caused by the low-pass component of PAN image, we propose to partition the PAN image and the spatial component of HS image into salient regions (i.e., image regions with rich details and no background redundancy) and nonsalient regions (i.e., image regions with background information) on which the detail extraction procedure is independently carried out. Note that, we only use the salient region of PAN image to improve the spatial quality of HS image, while its nonsalient region with background information does not play a role in HS pansharpening.
3) The detail extraction is performed on salient and nonsalient regions using a guided filter, whose goal is twofold: 1) the guided filter is a local image filter, which has the potential to minimize the overlap redundant information between the salient regions of PAN and HS images and 2) the guided filter plays a rectification role at nonsalient region with background redundancy such that the surrounding pixels belonging to the same object tend to have similar values in the pansharpened image.
4) The proposed detail extraction method is applied to four credible CS-based pansharpening methods, including the GIHS, Brovey, GS, and GSA approaches. Experimental results prove that the proposed scheme can improve the performance of these methods. Our method also has comparable or better performance than other state-of-the-art methods.

The remainder of this article is organized as follows. The CS methods, with related detail extraction techniques, are briefly reviewed in Section II. Section III elaborates on the details of the proposed algorithm and the implementation of the detail extraction procedure of the proposed method to the GIHS, Brovey, GS, and GSA methods. The experiment results are presented in Section IV. Section V summarizes this article.

II. Related Work

The CS-based pansharpening algorithms are popular for their simplicity and ease of implementation. The principle of the CS-based algorithms is to replace the spatial information of HS image with PAN image in a transform domain and then obtain high-resolution HS image by inverse transformation [4]. The greater the correlation between the replaced component and the PAN image, the better the fusion performance [26]. For the kth band of the HS image, their fusion steps can be summarized as follows:

$$\mathbf{H}_F^k = \widehat{\mathbf{H}}^k + \mathbf{G}^k \circ \mathbf{D} \tag{1}$$

where the superscript $k \in \{1, 2, \ldots, N\}$ is used to denote the kth band of the HS image, N is the number of bands in HS

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image, $\mathbf{H}_F$ is the pansharpened HS image, $\mathbf{H}$ represents the HS image, $\widehat{\mathbf{H}}$ represents the interpolated HS image upsampled to the same size as PAN image, $\circ$ is the symbol of element-by-element multiplication operation, $\mathbf{G}$ denotes the injection gain, and $\mathbf{D}$ is the extracted detail map. In CS-based methods, $\mathbf{D}$ is given by

$$\mathbf{D} = \mathbf{P} - \mathbf{I} \tag{2}$$

where $\mathbf{P}$ is the PAN image which is often referred to as the ideal detail image, and $\mathbf{I}$ denotes the so-called substituted spatial component of HS image, defined as

$$\mathbf{I} = \sum_{i=1}^{N} \omega_i \widehat{\mathbf{H}}^i \tag{3}$$

where the weights $\omega = [\omega_1, \omega_2, \ldots, \omega_N]^{\mathrm{T}}$ measure the spectral overlap between the bands of the HS image and the PAN image [26], [27].

The CS-based pansharpening algorithms have low complexity; hence they are popular in applications. The following is a brief review of several classic CS-based pansharpening algorithms, including the GIHS, Brovey, PCA, GS, and GSA pansharpening methods.

A. (G)IHS and Brovey

Tu *et al.* [14] extended the IHS algorithm and introduced the GIHS method to process the image with more than three bands. GIHS is a generalization of IHS, in which the image is transformed into the intensity (I), hue (H), and saturation (S) information to mimic the human visual system. The fusion step of GIHS can be summarized by (1), with the injection gain defined by

$$\mathbf{G}^k = \left(\sum_{i=1}^{N} \omega_i \right)^{-1}. \tag{4}$$

Equation (1) becomes

$$\mathbf{H}_F^k = \widehat{\mathbf{H}}^k + \left(\sum_{i=1}^{N} \omega_i \right)^{-1} (\mathbf{P} - \mathbf{I}) \tag{5}$$

where $\mathbf{I}$ is defined by (3). Generally, weight $\{\omega_i\}_{i=1,\ldots,N}$ equals to $1/N$. The weight can be optimized by the spectral response of HS and PAN. It can also be set to any arbitrary set of nonnegative values, and their sum may not be 1.

The widely known Brovey [15] pansharpening process can also be described by the general model in (1) using the injection gain

$$\mathbf{G}^k = \frac{\widehat{\mathbf{H}}^k}{\mathbf{I}} \tag{6}$$

which yields

$$\mathbf{H}_F^k = \widehat{\mathbf{H}}^k + \frac{\widehat{\mathbf{H}}^k}{\mathbf{I}} (\mathbf{P} - \mathbf{I}). \tag{7}$$

The injection gain in the Brovey pan-sharpening method is defined as the ratio between the interpolated HS image and the intensity component in order to maintain the local intensity contrast of the original image in the pansharpened HS image.

B. PCA

The PCA transformation is a commonly used dimensionality reduction technique. The idea of PCA is to map n-dimensional features to m-dimensions (m < n), which are new orthogonal features called principal components. The basic hypothesis of applying PCA to pansharpening is that after PCA transformation, the spatial information of HS images is concentrated in the first principal component, while spectral information is concentrated in other principal components. The gain matrices $\mathbf{G}$ and vector ω in (1) and (3) are obtained via the PCA procedure applied to the HS images.

C. GS and GSA

As a special case of PCA, the GS and GSA methods rely upon the projection of the high spectral resolution HS image into a new space to extract the spatial information, then replace this component with the PAN image, and finally obtain the pansharpened image by inverse transform. The pansharpening process of GS and GSA is described by (1), with the gain matrix given by

$$\mathbf{G}^k = \frac{\mathrm{cov}(\widehat{\mathbf{H}}^k, \mathbf{I})}{\mathrm{var}(\mathbf{I})} \tag{8}$$

where $\mathrm{cov}(\widehat{\mathbf{H}}^k, \mathbf{I})$ represents the covariance operation of $\widehat{\mathbf{H}}^k$ and $\mathbf{I}$, and $\mathrm{var}(\mathbf{I})$ denotes the variance operation of $\mathbf{I}$. For GS method, the weight in (3) is set to $\omega_i = 1/N$ [16]. As an improved version of GS, the GSA [17] calculates the weight by minimizing the mean square error between the low-passed filtered and downsampled PAN image and the estimated component, which is expressed as

$$\{\omega_1, \omega_2, \ldots, \omega_N\} = \mathrm{argmin} \left\| \sum_{i=1}^{N} \omega_i \mathbf{H}^i - \mathbf{P}_f \right\|^2 \tag{9}$$

where $\mathbf{P}_f$ is the filtered and downsampled PAN image. Therefore, the GS and GSA can be summarized as follows:

$$\mathbf{H}_F^k = \widehat{\mathbf{H}}^k + \frac{\mathrm{cov}(\widehat{\mathbf{H}}^k, \mathbf{I})}{\mathrm{var}(\mathbf{I})} (\mathbf{P} - \mathbf{I}). \tag{10}$$

The aforementioned pansharpening algorithms can be represented by a unified fusion model (1), that is, the whole pansharpening process can be decomposed into two successive processes of detail extraction and detail injection, wherein the detail extraction is represented by (2). According to (2), the principle of the CS-based pansharpening algorithms is to replace the spatial component of the HS image with the PAN image. Without considering the detail injection, it can be concluded from (5), (7), and (10) that they all use the PAN image as the ideal detail map. The unique distinction in detail extraction step among these algorithms is how to estimate the spectral weights used to linearly generate the spatial component of HS image.

III. Proposed Method

As discussed before, the low-frequency component of the PAN image and the mismatched spectral range between the PAN image and the HS image are the two most important

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causes to lead to spectral distortion in traditional CS-based pansharpening methods. To overcome these drawbacks, a new detail extraction scheme for HS pansharpening is proposed in this article. In this section, we first introduce the detail extraction scheme, and then apply it to four credible CS-based pansharpening methods.

A. Improved Detail Extraction

1) Preprocessing: To enhance the spatial information of the PAN image, a preprocessing which is called the Laplacian of Gaussian (LOG) enhancement algorithm is applied to the PAN image. The LOG algorithm can also reduce the noise of the PAN image in the simulation experiment, in which, the PAN image is obtained by averaging the visible light band of the HS image. The principle of the LOG image enhancement algorithm is that the PAN images are denoised by Gaussian convolution filtering and then the spatial details of the denoised PAN image are enhanced by the Laplace operator. This process can be expressed by the following formula:

$$\mathbf{P}_e = \mathbf{P} + \beta\ [\mathbf{P} * f_g(x, y)] \tag{11}$$

where $\mathbf{P}_e$ denotes the enhanced PAN image, $*$ represents the convolution operator, β is a constant determined by the center coefficient of the kernel of $f_g(x, y)$, and $f_g(x, y)$ denotes the kernel function of the LOG operator, which is defined as

$$f_g(x, y) = \frac{\partial^2}{\partial x^2} f(x, y) + \frac{\partial^2}{\partial y^2} f(x, y) \tag{12}$$

where $f(x, y)$ is the Gaussian convolution function.

2) Saliency Map Extraction by Spectral Residual (SR) Approach: The low-frequency component of the PAN image in traditional CS-based pansharpening algorithms always destroys the integrity and clarity of the pansharpened images, leading to loss of details, blurring of edges and other defects. The salient regions of the image preserve the spatial details, and nonsalient regions mainly contain the background information. Therefore, in this article, only the salient regions of the PAN image with rich details are used for pansharpening to reduce the spectral distortion of the pansharpened image caused by the low-frequency component of the PAN image. Partitioning the image into salient regions and nonsalient regions by binary segmentation and applying different fusion algorithms to different regions is an effective way to achieve this goal. It is known that the innovation areas of an image can be captured by the saliency map. Therefore, the salient region and nonsalient region of PAN and HS images can be obtained by binary segmentation of the saliency map. Furthermore, in order to reduce spectral distortion caused by the mismatched spectral range between the PAN image and the HS image, the proposed method utilizes the total saliency map composed by both PAN and HS images for binary segmentation. The following is a detailed description of generating the total saliency map.

There are many ways to extract the saliency map [35]–[37], and we use the SR method [35] in this article. The saliency maps of HS and PAN images are extracted by performing the SR method on the intensity component of the HS image $\mathbf{I}$ and the enhanced PAN image $\mathbf{P}_e$.

Some statistical properties of the image are transform-invariant, that is, when the image is transformed from the original spatial coordinates to the frequency coordinate system, these statistical properties of the image are still retained. The first step of the SR method is to perform the Fourier transform on the PAN image $\mathbf{P}_e$

$$\begin{aligned} F(u, v) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \mathbf{P}_e(x, y) e^{-2i\pi(ux+vy)} dx dy \\ &= A_P(u, v) \exp\{i B_P(u, v)\} \end{aligned} \tag{13}$$

where $A_P(u, v)$ is the amplitude spectrum, and $B_P(u, v)$ denotes the phase spectrum of the image. Since the log curve satisfies the local linear condition, we take the logarithm of $A_P(u, v)$ to yield the log spectrum $L(u, v)$ and use the local average filter to smooth $L(u, v)$ to obtain the average spectrum $\zeta(u, v)$. The SR $R_P(u, v)$ of the image $\mathbf{P}_e$ which describes the innovation part in a PAN image can be obtained by subtracting $\zeta(u, v)$ from $L(u, v)$

$$R_P(u, v) = L(u, v) - \zeta(u, v). \tag{14}$$

Finally, the saliency map $\mathbf{S}_P(x, y)$ representing the saliency of each pixel of the PAN image is reconstructed exactly by applying the inverse Fourier transform to the SR $R_P(u, v)$ and the phase spectrum $B_P(u, v)$

$$\mathbf{S}_P(x, y) = |F^{-1}[\exp\{R_P(u, v) + i B_P(u, v)\}]|^2. \tag{15}$$

Similarly, we use the SR approach to extract the saliency map of $\mathbf{I}$. Let $\mathbf{S}_H$ represent the saliency map of $\mathbf{I}$. The saliency map $\mathbf{S}_P$ is added with the saliency map $\mathbf{S}_H$ to obtain the total saliency map $\mathbf{S}$, and $\mathbf{S}$ represents the saliency areas with details of both PAN and HS images

$$\mathbf{S}(x, y) = \mathbf{S}_P(x, y) + \mathbf{S}_H(x, y). \tag{16}$$

3) Region Segmentation: The threshold plays an important role in the binary segmentation of the saliency map. Based on the global threshold, the Otsu method [31] is a powerful and popular thresholding technique, but this kind of method may give poor performance when processing images that contain dark background. Selecting the basic statistical distribution feature mean as the threshold is also a simple yet effective method. However, it relies heavily on having the evenly distributed grayscale values which are not always possible. In this article, we propose an effective threshold selection method to partition salient regions and nonsalient regions. The Gaussian mixture model (GMM) [32] is a linear combination of multiple Gaussian distribution functions, which can approximate any distribution, divide an image into several clusters, and use multiple Gaussian distributions to characterize the image. The reasonable statistical parameters for image description can be obtained by approximating $\mathbf{S}$ with the GMM. Inspired by this observation, an automatic threshold selection algorithm using GMM-based mean is developed in this article. The following details the threshold selection algorithm.

Since the GMM can represent any distribution, $\mathbf{S}$ follows GMM. Another way to think about it is, let $\mathbf{S}_H$ and $\mathbf{S}_P$ be modeled by the GMM. Since the Gaussian distribution has the linear transformation property, the sum of two independent

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random variables which follow the Gaussian distribution still obeys Gaussian distribution [33]. Thus, $\mathbf{S}$ obeys the GMM distribution. The GMM form of $\mathbf{S}$ is described as follows:

$$P(\mathbf{S}) \sim \sum_{r=1}^{K} \pi_r \mathcal{N}(s_c|\mu_r, \sigma_r), \quad \text{s.t } .\pi_r \geq 0, \ \sum_{r=1}^{K} \pi_r = 1 \tag{17}$$

where $c = 1, 2, \ldots, M$, M is the number of pixels, K denotes the number of Gaussian distributions, π_r is the probability that the rth component is selected, $\mathcal{N}(s_c|\mu_r, \sigma_r)$ denotes the Gaussian probability density function of the rth component in the GMM

$$\mathcal{N}(s_c|\mu_r, \sigma_r) = \frac{1}{\sqrt{2\pi}\sigma_r} \exp\ [-\frac{(s_c - \mu_r)^2}{2\sigma_r}] \tag{18}$$

where μ_r and σ_r denote the mean and variance, respectively.

Expectation maximization (EM) algorithm [34], [35] is one of the most popular tools to estimate the parameters $\{\pi_r, \mu_r, \sigma_r\}_{r=1,2,\ldots,K}$. The EM algorithm estimates these parameters by maximizing the log-likelihood function. It mainly contains E step and M step. A high order latent variable z_{cr} is assumed in the E step, with $z_{cr} \in \{0, 1\}$ and $\sum_{r=1}^{K} z_{cr} = 1$, presenting the assigned value of each GMM distribution. The posterior probability of the rth Gaussian mixture $\gamma(z_{cr})$ can be calculated as follows:

$$\gamma(z_{cr}) = \frac{\pi_r \mathcal{N}(s_c|\mu_r, \sigma_r)}{\sum_{r=1}^{K} \pi_r \mathcal{N}(s_c|\mu_r, \sigma_r)}. \tag{19}$$

The M step calculates μ_r, σ_r, and π_r by using $\gamma(z_{cr})$ obtained in the E step and also calculates the logarithmic likelihood function to distinguish whether the parameters or the log likelihood function are converged or not. The parameters $\{\pi_r, \mu_r, \sigma_r\}_{r=1,2,\ldots,K}$ yield the following estimated solutions:

$$\begin{aligned} \mu_r &= \frac{1}{N_r} \sum_{c=1}^{M} \gamma(z_{cr}) s_c \\ \sigma_r &= \frac{1}{N_r} \sum_{c=1}^{M} \gamma(z_{cr})(s_c - \mu_r)^2 \\ \pi_r &= \frac{N_r}{M} \end{aligned} \tag{20}$$

where $N_r = \sum_{c=1}^{M} \gamma(z_{cr})$. The E and M steps are repeated until these parameters converge or the log-likelihood function converges.

In the estimated parameters, μ_r represents the mean of the rth Gaussian component, and π_r denotes the probability of the occurrence of the rth Gaussian component. Therefore, $\sum_{r=1}^{K} \pi_r \mu_r$ represents the mean of the mixed Gaussian components of the total saliency map. The mean is an effective parameter to measure the intensity. In this article, we use $\sum_{r=1}^{K} \pi_r \mu_r$ to perform threshold-based segmentation on the total salient map to obtain the binary salient and nonsalient regions. This process can be described by

$$\mathbf{W}(x, y) = \begin{cases} 1, & \mathbf{S}(x, y) \geq \sum_{r=1}^{K} \pi_r \mu_r \\ 0, & \text{otherwise} \end{cases} \tag{21}$$

where $\mathbf{W}$ is the binary map, $\mathbf{W}(x, y)$ and $\mathbf{S}(x, y)$ denote the pixel in the xth row and the yth column of $\mathbf{W}$ and $\mathbf{S}$, respectively.

Once the binary map of the total saliency map is adaptively obtained, the salient regions of $\mathbf{P}_e$ and $\mathbf{I}$ are constructed as

$$\mathbf{P}_S = \mathbf{P}_e \circ \mathbf{W} \tag{22}$$

$$\mathbf{H}_S = \mathbf{I} \circ \mathbf{W} \tag{23}$$

where $\mathbf{P}_S$ is the salient region of the enhanced PAN image, $\mathbf{H}_S$ denotes the salient region of the spatial component of HS image, and $\circ$ is the element-wise multiplication operator. The nonsalient regions of $\mathbf{P}_e$ and $\mathbf{I}$ are easily constructed by subtracting salient region maps from the original images

$$\mathbf{P}_{\mathrm{NS}} = \mathbf{P}_e - \mathbf{P}_S = \mathbf{P}_e - \mathbf{P}_e \circ \mathbf{W} \tag{24}$$

$$\mathbf{H}_{\mathrm{NS}} = \mathbf{I} - \mathbf{H}_S = \mathbf{I} - \mathbf{I} \circ \mathbf{W} \tag{25}$$

where $\mathbf{P}_{\mathrm{NS}}$ represents the nonsalient region of $\mathbf{P}_e$, and $\mathbf{H}_{\mathrm{NS}}$ denotes the nonsalient region of $\mathbf{I}$.

4) Fusions of Salient Regions and Nonsalient Regions by Guided Filtering: For the salient region with rich details, we generate the detail map via fusing the salient region of the PAN image and that of the spatial component of the HS image. For the nonsalient region with background information and less details, the detail map is generated only from the nonsalient region of the HS image to avoid injecting the low-pass information of the PAN image.

The guided filter [36] is a local image filter, which has the potential to minimize the overlap redundancy information between the PAN and HS images. Therefore, for the salient regions with rich details, we propose a guided filter-based fusion method to obtain the detail map. The guided filtering is performed on $\mathbf{P}_S$ with $\mathbf{H}_S$ serving as the guidance image

$$\mathbf{G}_S = G(\mathbf{P}_S, \mathbf{H}_S) \tag{26}$$

where G represents the function of the guided filter, and $\mathbf{G}_S$ denotes the output image. According to the theory of guided filtering, the output image $\mathbf{G}_S$ is a linear transformation of the guidance image $\mathbf{H}_S$ in a small window o_h centered at the pixel h

$$\mathbf{G}_{S,c} = \frac{1}{|T|} \sum_{h \in o_c} a_h \mathbf{H}_{S,c} + \frac{1}{|T|} \sum_{h \in o_c} b_h, \quad \forall c \in o_h \tag{27}$$

where $c = 1, 2, \ldots, M$, $\mathbf{G}_{S,c}$ and $\mathbf{H}_{S,c}$ are the cth pixel of $\mathbf{G}_S$ and $\mathbf{H}_S$, o_c is a local window centered at the pixel c, $|T|$ is the number of pixels in o_c. a_h, and b_h are constant linear coefficients, which can be calculated by solving a linear ridge regression model

$$a_h = \frac{1/|T| \left(\sum_{c \in o_h} \mathbf{H}_{S,c} \mathbf{P}_{S,c} \right) - \xi_h \overline{\mathbf{P}}_{S,h}}{\theta_h^2 + \varsigma} \tag{28}$$

$$b_h = \overline{\mathbf{P}}_{S,h} - a_h \xi_h \tag{29}$$

where ς is a regularization parameter, ξ_h and θ_h^2 represent the mean and variance of the guidance image, and $\overline{\mathbf{P}}_{S,h}$ is the mean of the input image in o_h. The guided filter has two parameters which are the filter window size and the regularization parameter ς, and the values of these two parameters

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Algorithm 1 Region-Fusion-Based Detail Extraction Algorithm

Input: The PAN image $\mathbf{P}$, and the substituted detail component of the HS image $\mathbf{I}$
Output: The injection detail maps of the salient and nonsalient regions $\mathbf{D}_S$ and $\mathbf{D}_{\mathrm{NS}}$

Do the preprocessing (LOG enhancement algorithm) on the PAN image, yielding $\mathbf{P}_e$;
Obtain the total saliency map $\mathbf{S}$ by SR method shown in subsection-II-A-2;
Initialize the parameters $\{\pi_r, \mu_r, \sigma_r\}_{r=1,2,\ldots,K}$, and preset the number of mixture components K;
while log likelihood not converged **do**
  E Step: Evaluate the posterior responsibility $\gamma(z_{cr})$ according to (19) for $c = 1, 2, \ldots, M$, $r = 1, \ldots, K$;
  M Step: Evaluate the parameters $\{\pi_r, \mu_r, \sigma_r\}$ according to (20) for $r = 1, \ldots, K$;
Compute $\mathbf{W}$ according to (21);
Compute the salient regions and the nonsalient regions of $\mathbf{P}_e$ and $\mathbf{I}$ according to (22)-(23) and (24)-(25), respectively;
Generate the fused salient region map $\mathbf{F}_S$ and the fused nonsalient region map $\mathbf{F}_{\mathrm{NS}}$ according to (26)-(31);
Perform the histogram matching of $\mathbf{F}_S$ to the substituted component $\mathbf{H}_S$ and the histogram matching of $\mathbf{F}_{\mathrm{NS}}$ to $\mathbf{H}_{\mathrm{NS}}$;
Generate the injection detail map of the salient region $\mathbf{D}_S$ and the injection detail map of the nonsalient region $\mathbf{D}_{\mathrm{NS}}$ according to (32) and (33), respectively.

are generally determined empirically in practice. In this article, empirically, the window size and the regularization parameter are set to 15 and 10^{-6}, respectively. The structure details of the guidance image $\mathbf{H}_S$ are transferred to the output image $\mathbf{G}_S$. Thus, the difference image between $\mathbf{P}_S$ and $\mathbf{H}_S$ is obtained by subtracting the output image $\mathbf{G}_S$ from the $\mathbf{P}_S$, and the fused image of the salient regions is structured by adding the difference image with $\mathbf{H}_S$

$$\mathbf{F}_S = \mathbf{D}_S + \mathbf{H}_S = \mathbf{P}_S - \mathbf{G}_S + \mathbf{H}_S \tag{30}$$

where $\mathbf{F}_S$ denotes the ideal detail map for the salient region, and $\mathbf{D}_S$ is the difference image.

The nonsalient region with background redundancy might not be well-aligned with the boundaries of objects, which causes the block artifact in the fused image. Optimization-based approaches are the most popular methods to solve this problem, but it is inefficient to construct an energy function. We know that guided filtering [36] has the capability of eliminating block artifact and transferring spatial details. In order to avoid artifacts at the boundaries of the pansharpened image, a rectification step using guided filtering is performed on the nonsalient regions that have background redundancy and no spatial details. The rectification step with guided filtering can improve the performance of the pansharpened image such that adjacent pixels belonging to the same target tend to have similar values in the pansharpened image

$$\mathbf{F}_{\mathrm{NS}} = G(\mathbf{H}_{\mathrm{NS}}, \mathbf{H}_{\mathrm{NS}}) \tag{31}$$

where $\mathbf{F}_{\mathrm{NS}}$ denotes the ideal detail map for the nonsalient region.

Algorithm 2 CS-Based Pansharpening Algorithm Based on Region-Fusion Based Detail Extraction Algorithm

Input: The available PAN image $\mathbf{P}$ and the available HS image $\mathbf{H}$
Output: The pansharpened HS image $\mathbf{H}_F$

Upscale the available HS image $\mathbf{H}$ to the scale $\mathbf{P}$, yielding $\widehat{\mathbf{H}}$;
Generate the detail component of the HS image $\mathbf{I}$ using each CS-based method ;
Obtain the injection detail maps of the salient and nonsalient regions $\mathbf{D}_S$ and $\mathbf{D}_{\mathrm{NS}}$ through Algorithm 1 ;
for $k = 1, 2, \ldots, N$ **do**
  Calculate the injection gains of the salient and nonsalient regions $\mathbf{G}_S^k$ and $\mathbf{G}_{\mathrm{NS}}^k$ according to (34) and (35) for Brovey, and (36) and (37) for GS and GSA
Compute the fused HS image $\mathbf{H}_F^k$ according to $\mathbf{H}_F^k = \widehat{\mathbf{H}}^k + \mathbf{G}_S^k \circ \mathbf{D}_S + \mathbf{G}_{\mathrm{NS}}^k \circ \mathbf{D}_{\mathrm{NS}}$

Then the histogram matching of $\mathbf{F}_S$ to the substituted component $\mathbf{H}_S$ and the histogram matching of $\mathbf{F}_{\mathrm{NS}}$ to the substituted component $\mathbf{H}_{\mathrm{NS}}$ are, respectively, performed to obtain the histogram-matched results.

5) Generation of Injection Details: The detail map of traditional CS-based fusion algorithms is generated according to formula (2). They all regard the PAN image as the ideal spatial detail map. In other words, the ideal detail map, required for constructing the injection detail map $\mathbf{D} = \mathbf{P} - \mathbf{I}$, is directly the PAN image. However, spectral distortion often occurs due to the spectral range mismatch between HS and PAN images. Based on the idea of the CS method, we fuse the spatial components of the PAN image with that of the HS image and use the fused detail map as the ideal detail map to replace the spatial component of the interpolated HS image. In our improved version, for the salient regions, the injection detail map $\mathbf{D}_S$ can be obtained as follows:

$$\mathbf{D}_S = \mathbf{F}_S - \mathbf{H}_S \tag{32}$$

where $\mathbf{H}_S$ is the detail component of the HS image or the so-called substituted details. Similarly, for the nonsalient region, the injection detail map is expressed as

$$\mathbf{D}_{\mathrm{NS}} = \mathbf{F}_{\mathrm{NS}} - \mathbf{H}_{\mathrm{NS}} \tag{33}$$

where $\mathbf{D}_{\mathrm{NS}}$ is the injection detail map of the nonsalient regions, $\mathbf{F}_{\mathrm{NS}}$ and $\mathbf{H}_{\mathrm{NS}}$, respectively, denote the ideal detail map and the substituted details of the nonsalient regions. The proposed region-fusion based detail extraction algorithm is summarized in Algorithm 1.

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TABLE I
CHARACTERISTICS OF THE DATA SETS

Data set	Instrument	Bands	Spectral Range	Resolution (HS/PAN)	Dimensions (HS/PAN)
Moffett field	AVIRIS	176	$0.38-2.5\mu$m	100m / 20m	50×30 / 250×150
Hyperion	EO-I	174	$0.4-2.5\mu$m	30m / 10m	70×70 / 210×210

B. Regional Injection Gain Estimation: GIHS, BP, GS, and GSA Using Improved Detail Map Extraction

In this section, we will apply the proposed detail extraction algorithm to the GIHS, Brovey, GS, and GSA methods, and we will also define the injection gains of these four algorithms based on the image regions achieved via saliency analysis and GMM.

On the one hand, the GIHS, Brovey, GS, and GSA algorithms belong to CS class and can be described in (1). Injection detail maps in these four methods are obtained by substituting spatial detail components of HS images with PAN images, described by (2) and (3). In (2) and (3), $\mathbf{P}$ is the PAN image which is regarded as the ideal detail image, $\mathbf{I}$ denotes the so-called substituted details component, and the weights $\omega = [\omega_1, \omega_2, \ldots, \omega_N]^{\mathrm{T}}$ measure the spectral overlap between the bands of the HS image and PAN image. In general, the weights $\omega = [\omega_1, \omega_2, \ldots, \omega_N]^{\mathrm{T}}$ in GIHS, Brovey, and GS equal to $1/N$. For GSA, the weights are calculated by minimizing MSE with respect to a low-pass component of the PAN image and a weighted average of the HS bands. The detail extraction step proposed in Section III-A can be directly implemented to the GIHS, Brovey, GS, and GSA algorithms to replace (2) and (3), that is, the injection detail map $\mathbf{D}$ in (2) is replaced by the detail maps of the salient and nonsalient regions $\mathbf{D}_S$ and $\mathbf{D}_{\mathrm{NS}}$.

On the other hand, the injection gains of the GIHS and Brovey algorithms are defined as (4) and (6), respectively, whereas the injection gains of the GS and GSA algorithms are calculated by (8).

Since the definition of injection gain in the GIHS algorithm is independent of detail extraction, we only replace the detail extraction step of the GIHS algorithm and continue to use its injection gain. For the Brovey method, the injection gain is calculated according to (6). For applying the proposed method, the injection gain is easily calculated locally on the salient regions and the nonsalient regions. Formally, for the kth band, the injection gains of the salient and nonsalient regions $\mathbf{G}_S^k$ and $\mathbf{G}_{\mathrm{NS}}^k$ are, respectively, calculated as

$$\mathbf{G}_S^k = \frac{\widehat{\mathbf{H}}_S^k}{\mathbf{H}_S} \tag{34}$$

$$\mathbf{G}_{\mathrm{NS}}^k = \frac{\widehat{\mathbf{H}}_{\mathrm{NS}}^k}{\mathbf{H}_{\mathrm{NS}}} \tag{35}$$

where $k = 1, 2, \ldots, N$ denotes the kth band, $\widehat{\mathbf{H}}_S^k = \widehat{\mathbf{H}}^k \circ \mathbf{W}$ and $\widehat{\mathbf{H}}_{\mathrm{NS}}^k = \widehat{\mathbf{H}}^k - \widehat{\mathbf{H}}_S^k$ denote the salient and nonsalient regions of the interpolated HS image.

For GS and GSA methods, the injection gain is calculated by (8). According to (8), the injection gains of the salient and nonsalient regions $\mathbf{G}_S^k$ and $\mathbf{G}_{\mathrm{NS}}^k$ are, respectively, defined as

$$\mathbf{G}_S^k = \frac{\mathrm{cov}\left(\widehat{\mathbf{H}}_S^k, \mathbf{H}_S\right)}{\mathrm{var}(\mathbf{H}_S)} \tag{36}$$

$$\mathbf{G}_{\mathrm{NS}}^k = \frac{\mathrm{cov}\left(\widehat{\mathbf{H}}_{\mathrm{NS}}^k, \mathbf{H}_{\mathrm{NS}}\right)}{\mathrm{var}(\mathbf{H}_{\mathrm{NS}})}. \tag{37}$$

The proposed region-pansharpening method can be divided into three consecutive steps: 1) extracting the injection detail map $\mathbf{D}$ by the proposed detail extraction algorithm; 2) evaluating the injection gain; and 3) generating the fused HS image according to model (1). The CS-based HS pansharpening algorithm based on the proposed detail extraction strategy is summarized in Algorithm 2.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Data Sets

To compare the fusion performance of different pansharpening methods, both synthetic and real data sets are considered in the experiment. These experimental data sets provided by different sensors consist of different characteristics and heterogeneous structures. For clarity, the properties of the data sets used in the experiment are summarized in Table I.

The first experiment was conducted on the Moffett field data set, which is a synthetic data set. This data set was collected by the airborne HS instrument Airborne Visible Infrared Image Spectrometer (AVIRIS). The Moffett field data set shows a mixed scene of rural and urban areas. It contains 224 bands in the spectral range of 0.38–2.5 μm, yielding a complete VIS-NIR-SWIR spectrum. The noise bands and water vapor absorption bands were removed, and the remaining 176 bands were used for experiments. The dimension of the HS image is 50×30 pixels with the spatial resolution of 20 m, whereas the size of the PAN image is 250×150 with a spatial resolution of 100 m. For the synthetic data set, HS images are available, whereas PAN images are not available. The available HS images serve as the reference image. The HS images and PAN images used for pansharpening are generated by Wald's protocol [37]. Specifically, the synthetic HS images are generated by performing a Gaussian blurring to the high spatial resolution reference HS image with a $9 * 9$ Gaussian blur, and downsampling every c pixels (c is the downsampling ratio) in both the length and width modes of the result of the Gaussian blurring step. The synthetic PAN images are generated by the pansharpening spectral response that averages the bands of the visible range.

The second and third experiments were carried out on the Hyperion data set which is a real data set. This data set was collected by the National Aeronautics and Space Administration EO-1 satellite which was launched on

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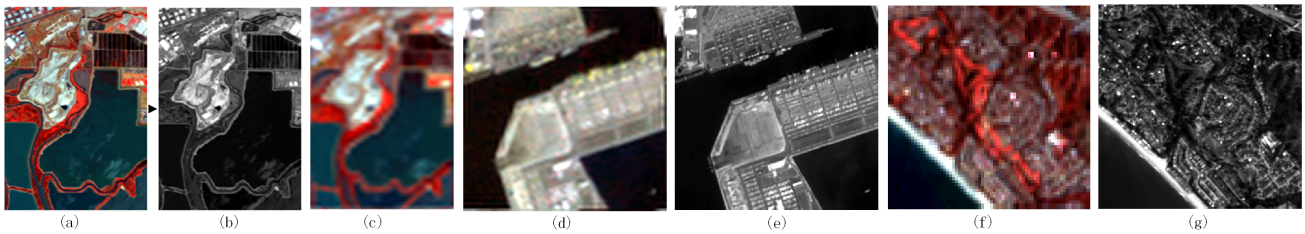


Fig. 1. Data sets used in the experiments. (a) Reference HS image of Moffett field data set. (b) Synthetic PAN image of Moffett field data set. (c) Synthetic HS image of Moffett field data set (it is upscaled to the PAN size). (d) and (e) Two sets of real HS image of Hyperion data set. (They are upscaled to the PAN size.) (f) and (g) Two sets of real PAN image of Hyperion data set.

November 21, 2000. The EO-1 spacecraft carried three primary instruments that are the Advanced Land Imager (ALI), the Hyperion, and the Linear Etalon Imaging Spectrometer Array (LEISA) Atmospheric Corrector (LAC). Operating in a pushbroom fashion at an orbit of 705 km, the ALI instrument is able to collect the Landsat-type PAN image with 10-m spatial resolution and the Landsat type MS image with a spatial resolution of 30 m. The Hyperion instrument provides the HS image for improved Earth surface characterization. The HS image provided by Hyperion contains 220 bands in the spectral range of 0.4–2.5 μm. After removing the invalid band, the remaining 174 bands are used for the experiments. The dimensions of the PAN and HS images are 210×210 and 70×70, respectively.

B. Compared Algorithms

To verify the advantages of the proposed algorithm, four algorithms, namely GIHS [14], Brovey [15], GS [16], and GSA [17], are selected from the CS class, and have been employed for comparison. The proposed algorithm is applied to these four CS-based methods and is named as GIHS_proposed, Brovey_proposed, GS_proposed, and GSA_proposed, respectively. Several state-of-the-art algorithms including GFPCA [38], Bayesian Naïve [6], Bayesian sparse [8], AWLP [39], SEGM [25], and STF [40] that represent the recent developments of HS pansharpening are also used for comparison. Among these compared methods, the CS-based methods are known for their simplicity and ease of implementation. The AWLP method is an MRA-based method and won the first prize of the IEEE Data Fusion Contest in 2006. The GFPCA method is a popular hybrid method, which combines the advantages of the guided filter and PCA, and overcomes the shortcomings of the PCA algorithm by using the properties of smooth filtering and edge-preserving characteristics of the guided filter. The Bayesian Naïve and Bayesian sparse methods are the representative algorithms of the Bayesian-based pansharpening algorithms, which provide a good balance in spatial sharpness and spectral fidelity. The SEGM method is a local-based pansharpening algorithm and achieves excellent fusion performance. In the experiments, the SEGM method that applies the binary partition tree image segmentation to the GSA method is adopted for comparison. These compared state-of-the-art methods include the latest effective works and cover various categories of the existing fusion algorithms. Therefore, these algorithms are selected in this article for comparison with the proposed algorithm.

C. Quality Assessment

In order to prove the effectiveness of the proposed method, the experiments are designed from the perspectives of subjective qualitative analysis and the objective quantitative analysis respectively. One synthetic data set and two real data sets are tested in the experiments, and the quality assessments for them are described in detail as follows.

1) Quality Assessment of Synthetic Data: For the synthetic data set, the HS images are available and the PAN images are not available. The available HS image shown in Fig. 1(a) serves as a reference image. Fig. 1(b) and (c) shows the synthetic PAN and the synthetic HS image, respectively. The synthetic HS and PAN images are fused by each method to obtain the fused HS images, and the fused HS images are compared with the reference HS image to evaluate the subjective performance and the quantitative quality measures. Therefore, the fusion results of the synthetic data set should be evaluated at reduced resolution. Seven widely used indexes, including cross correlation (CC) [41], spectral angle mapper (SAM), erreur relative globale adimensionnelle de synthse (ERGAS) [37], and universal image quality index (Q) [42], [43] are used to analyze the performance of different pansharpening methods quantitatively. Note that CC is calculated by averaging the CC values of each band and assesses the spatial similarity between the results and the reference images. SAM measures the degree of distortion between the spectral vector of the pansharpened results and that of the reference image. ERGAS is a global index used to evaluate the spectral preservation and spatial quality of fused images. Q measures contrast distortion, luminance distortion, and loss of correlation, and also measures the spectral distortion. For the CC and Q indexes, the ideal values are 1. The smaller the values of SAM and ERGAS, the better the performance of the fused images.

2) Quality Assessment of Real Data: For the real data set, the low spatial resolution HS image and the high spatial resolution PAN image are available. Fig. 1(d)–(g) shows the HS and PAN images of the two experimental data sets. The available HS and PAN images are fused by using the compared methods, and the subjective performance is evaluated by comparing each fused HS image. To assess the objective quantitative quality, for the real data set, two assessment procedures are adopted, namely the reduced-resolution assessment and the full-resolution validation [3].

1) *Reduced-Resolution Assessment:* The reduced-resolution assessment is based on Wald's protocol [37].

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Fig. 2. Pansharpening results of Moffett field data. (a) Reference HS image. (b) GFPCA. (c) Bayesian Naïve. (d) Bayesian sparse. (e) AWLP. (f) STF. (g) SEGM. (h) GIHS_original. (i) GIHS_proposed. (j) Brovey_original. (k) Brovey_proposed. (l) GS_original. (m) GS_proposed. (n) GSA_original. (o) GSA_proposed.

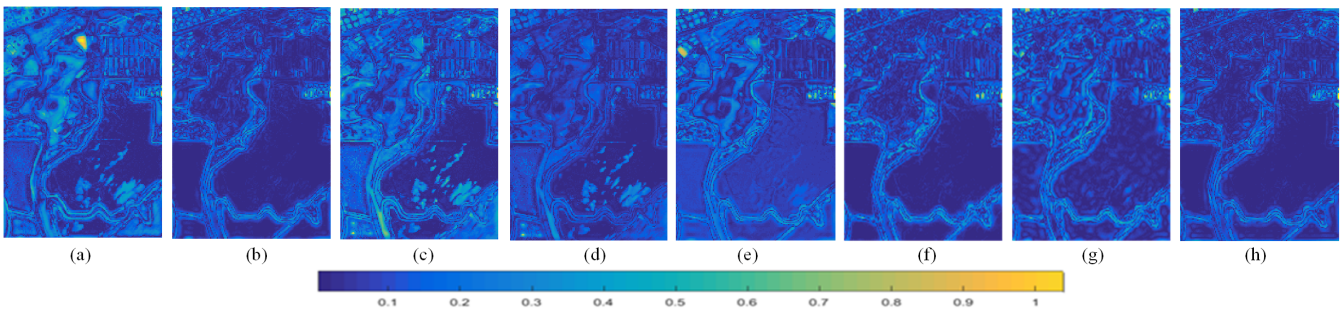


Fig. 3. Error images of results obtained by different pansharpening methods (Moffett field data set). (a) GIHS_original. (b) GIHS_proposed. (c) Brovey_original. (d) Brovey_proposed. (e) GS_original. (f) GS_proposed. (g) GSA_original. (h) GSA_proposed.

The original real HS image serves as the reference image, and the reduced-resolution HS and PAN images generated by degrading the available real HS and PAN images are fused to obtain the pansharpened HS image. The pansharpened HS image is compared with the original HS image to accomplish the reduced-resolution objective quality assessment. The indexes, i.e., CC, SAM, ERGAS, and Q, are utilized to quantify the fusion performance.

2) *Full-Resolution Validation:* The full-resolution validation is performed on the original PAN and HS images to assess the quality evaluation at the original resolution of the data. The quality with no reference (QNR) image [44] index is adopted. The QNR is a full resolution assessment index and is calculated by two values D_λ and D_s, where D_λ measures spectral distortion, and D_s measures spatial distortion. When both D_λ and D_s are 0, QNR achieves the optimal value of 1.

D. Experimental Results of Synthetic Data

The fused results of different pansharpening algorithms on the Moffett field date are displayed in Fig. 2. Fig. 2(a) presents the available high-resolution HS image which serves as the reference image. Fig. 2(b)–(g) displays the pansharpened results of several considered algorithms, including GFPCA,

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TABLE II
QUANTITATIVE ASSESSMENT FOR THE EXPERIMENTAL SYNTHETIC DATA SETS

Data set		Moffett Field				Hyperion1				Hyperion2			
Index		CC	SAM	ERGAS	Q	CC	SAM	ERGAS	Q	CC	SAM	ERGAS	Q
Reference		1	0	0	1	1	0	0	1	1	0	0	1
GFPCA		0.9283	12.1586	6.7014	0.8001	0.9397	14.8244	13.388	0.6492	0.7361	6.2363	11.1183	0.8298
Bayesian Naïve		0.9422	7.4219	5.7655	0.8792	0.9759	11.7625	7.8845	0.8314	0.8679	4.2968	7.7966	0.9384
Bayesian Sparse		0.9581	7.8956	5.0038	0.8692	0.9744	14.2641	8.7522	0.8029	0.8113	5.7077	9.3469	0.9205
AWLP		0.9655	7.3254	4.5083	0.8805	0.9666	9.6678	9.3173	0.8330	0.8715	3.9914	7.6797	0.9163
STF		0.9640	7.3223	4.7785	0.8788	0.9762	8.8621	8.7338	0.8598	0.8661	4.0012	7.6489	0.9212
SEGM		0.9685	8.1498	**4.3343**	**0.8892**	0.9781	9.6163	7.7438	0.8674	0.8947	3.8989	6.6916	0.9393
GIHS	original	0.9199	11.5947	6.9774	0.8377	0.9315	9.9075	13.3233	0.8550	0.7245	3.9972	12.8415	0.9375
	proposed	0.9602	7.4471	4.9217	0.8833	0.9700	8.8135	8.8966	0.8770	0.8007	3.6251	9.7736	0.9475
Brovey	original	0.9010	7.3811	7.0062	0.7841	0.9307	8.4842	13.3743	0.8317	0.6981	**3.2220**	11.7742	0.9238
	proposed	0.9600	7.3223	4.9781	0.8817	0.9717	**8.4839**	8.8816	0.8734	0.8023	3.3597	9.1223	0.9411
GS	original	0.9299	12.4823	6.6839	0.8237	0.9383	10.1882	13.1434	0.8391	0.6879	4.2041	11.3998	0.8652
	proposed	0.9602	7.4269	4.8979	0.8844	0.9818	8.8523	7.1848	0.8786	0.8981	3.4009	6.3359	0.9533
GSA	original	0.9554	10.0227	5.3507	0.8550	0.9618	10.4465	10.3904	0.8434	0.8975	3.3374	6.4423	0.9539
	proposed	**0.9695**	**7.2432**	4.5203	0.8866	**0.9843**	8.5175	**6.7365**	**0.8820**	**0.9099**	3.2358	**6.1196**	**0.9545**

Bayesian Naïve, Bayesian sparse, AWLP, STF, and SEGM methods. In Fig. 2(h), (j), (l), and (n), we show the pansharpened results of GIHS, Brovey, GS, and GSA, respectively. Fig. 2(i), (k), (m), and (o) shows their improved pansharpened results obtained by employing the detail extraction step proposed in this article. A visual inspection of the pansharpened results shows that most of the pansharpening algorithms achieve good performance in rendering the spatial details, except for the GFPCA method. In order to avoid excessive modification of spectral information, the amount of details added to the HS image in GFPCA controlled by the guided filter may be insufficient. Fig. 2(b) shows the worst spatial and spectral quality compared to the others. SEGM is a context-adaptive pan-sharpening method based on image segmentation. The experimental result in Fig. 2(g) shows that the SEGM algorithm has high spectral fidelity on the Moffett field data set, but the image edge is blurred due to insufficient spatial detail injection. Overall, the Bayesian Naïve, Bayesian sparse, AWLP, and STF methods provide a good tradeoff between spectral fidelity and sharpness.

This article is focused on comparing the performance of two groups of four CS-based pansharpening algorithms including GIHS, Brovey, GS, and GSA. The spectral distortion involved by the GIHS and Brovey methods and, to a small extent, by the GS method are particularly visible because small differences in color can be seen in Fig. 2(h), (j) and (l) with respect to the reference image. Compared with the GS method, the GSA method has a certain improvement in spectral fidelity, but the spatial quality is still not ideal. The spectral and spatial improvements by employing the proposed detail extraction approach instead of the original ones can be easily observed. By comparing the results of the GIHS, Brovey, GS, and GSA algorithms with their improved versions, one can note that the detail extraction procedure proposed in this article is particularly beneficial, leading to a superior performance, especially in spatial details, for all four pansharpening algorithms. In addition, the proposed method has comparable overall spatial performance compared to several currently advanced algorithms. The experimental results on the Moffett field data set show that the proposed algorithm can reduce the spectral distortion of the original HS image and still maintain its edge detail and improve the spatial resolution.

In order to further analyze the performance of different methods qualitatively, we obtain the error images by calculating the pixel difference between the fused image obtained by each pansharpening method and the reference image. The deficiencies of different pansharpening methods can be seen more clearly by the error images. Fig. 3 presents the error images of different pansharpening methods on the Moffett field data set, where small differences are represented in blue, and large differences are in green and yellow. It can be seen that the error images of the GIHS and Brovey methods are particularly prominent on the edge texture because of the insufficient spatial information. The blue regions in the error images of the proposed CS-based methods are larger than the blue regions in the error images of the original CS-based methods. This means that the spatial and spectral distortion is reduced by the proposed algorithm. This further proves that the CS-based algorithms using the detail extraction procedure proposed in this article provide better results than the original methods.

The quality metric values of different pansharpening algorithms on the Moffett field data sets are listed in Table II. For clarity, the best value for each index is marked in bold and the second-best value is underlined. It can be noticed that the proposed method provides almost all the best values for the quality metric indexes on the Moffett field data sets. In contrast, GFPCA has the worst performance for both qualitative and quantitative analysis. The SEGM method obtains the second largest value of CC, the largest Q, and the smallest values of ERGAS. Consistent with the qualitative analysis, the quality indexes of the GSA method have a significant improvement compared with that of the GS method. Compared with the original algorithms, the performance of the

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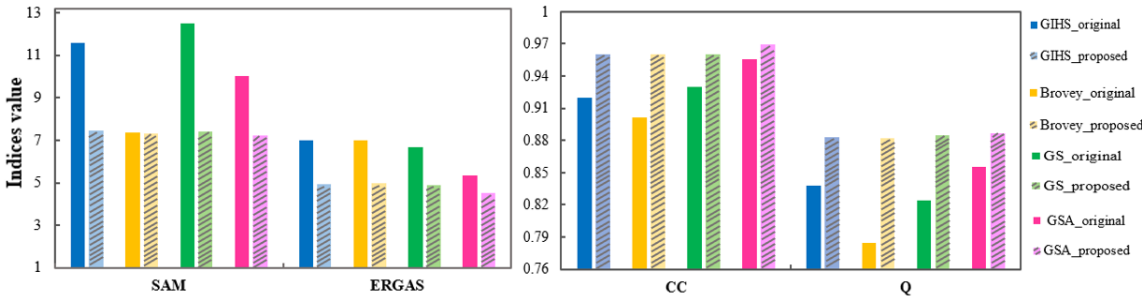


Fig. 4. Quantitative assessment for Moffett field data.

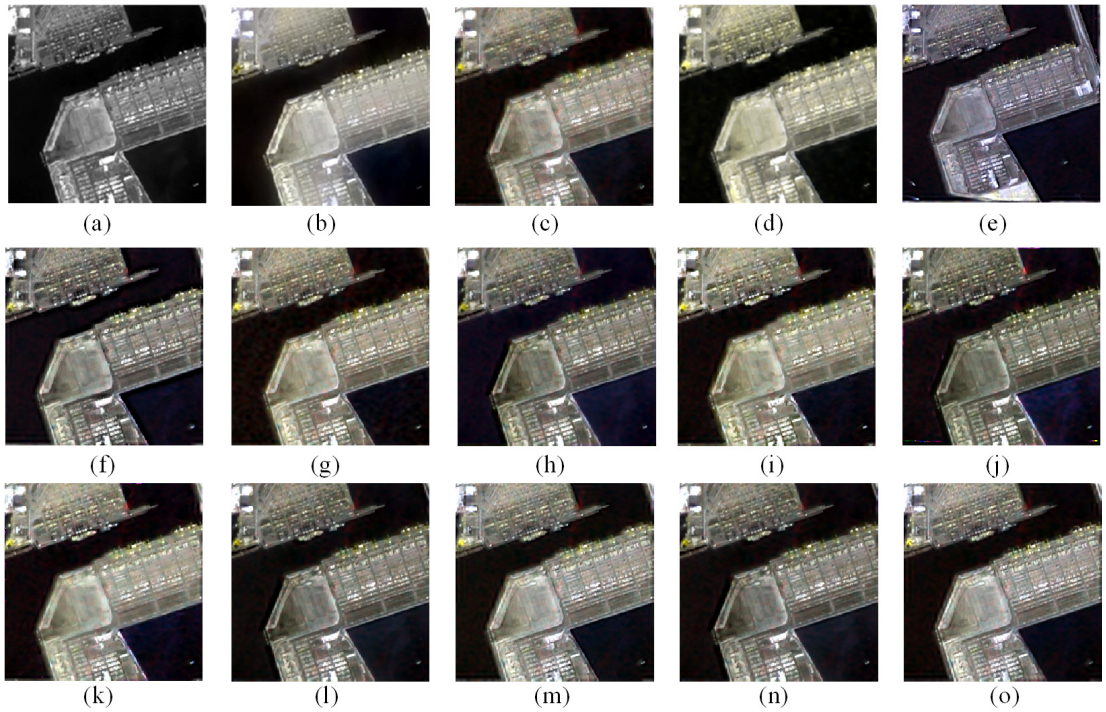


Fig. 5. Pansharpening results on Hyperion1 data. (a) Real PAN image. (b) GFPCA. (c) Bayesian Naïve. (d) Bayesian sparse. (e) AWLP. (f) STF. (g) SEGM. (h) GIHS_original. (i) GIHS_proposed. (j) Brovey_original. (k) Brovey_proposed. (l) GS_original. (m) GS_proposed. (n) GSA_original. (o) GSA_proposed.

GIHS, Brovey, GS, and GSA algorithms that apply the detail extraction step proposed in this article is greatly improved. In particular, the improved GSA algorithm obtains the optimal CC and SAM values, and the Q value is second only to that of the SEGM algorithm. In general, the performance of the improved GIHS and GSA algorithms is not only superior to that of the original algorithms but also better than that of several advanced pansharpening algorithms. The objective performance of the improved GSA algorithm is comparable to that of SEGM, which cannot be realized by the original GSA algorithm. The CC values of the proposed GIHS, Brovey, GS, and GSA algorithms are greatly improved compared to the original algorithms and the improved GSA algorithm provides the largest CC value, confirming its state-of-the-art performance in spatial quality. Fig. 4 shows the bar chart comparing the objective performance of the original GIHS, Brovey, GS, and GSA algorithms and their improved versions. It is more intuitive to show that the spatial detail extraction step proposed in this article effectively improves the performance of the fused image.

E. Experimental Results of Real Data

In order to verify the effect of the proposed method on the real data set, the second and the third experiments are performed on the Hyperion data sets. The original PAN images, as well as the results of different pansharpening methods on hyperion1 and hyperion2 data sets are shown in Figs. 5 and 6, respectively. It can be seen that all the pansharpening methods can provide fused results with good visual effect. GFPCA generates severe spatial distortion. This is visible in both Figs. 5(b) and 6(b), where the edges and textures of the objects are blurred compared to the reference PAN image. The images produced by the GFPCA method look fuzzy due to insufficient details. The original CS-based pansharpening methods often produce a severe spectral distortion. As demonstrated in Fig. 6, the results of the original GIHS, Brovey, and GS methods suffer from varying degrees of spectral distortion. The AWLP, STF, and SEGM methods yield fused results with excellent spectral fidelity. Only in some building areas, the original CS-based pansharpening algorithms provide higher performance than the Bayesian Naïve, AWLP, Bayesian

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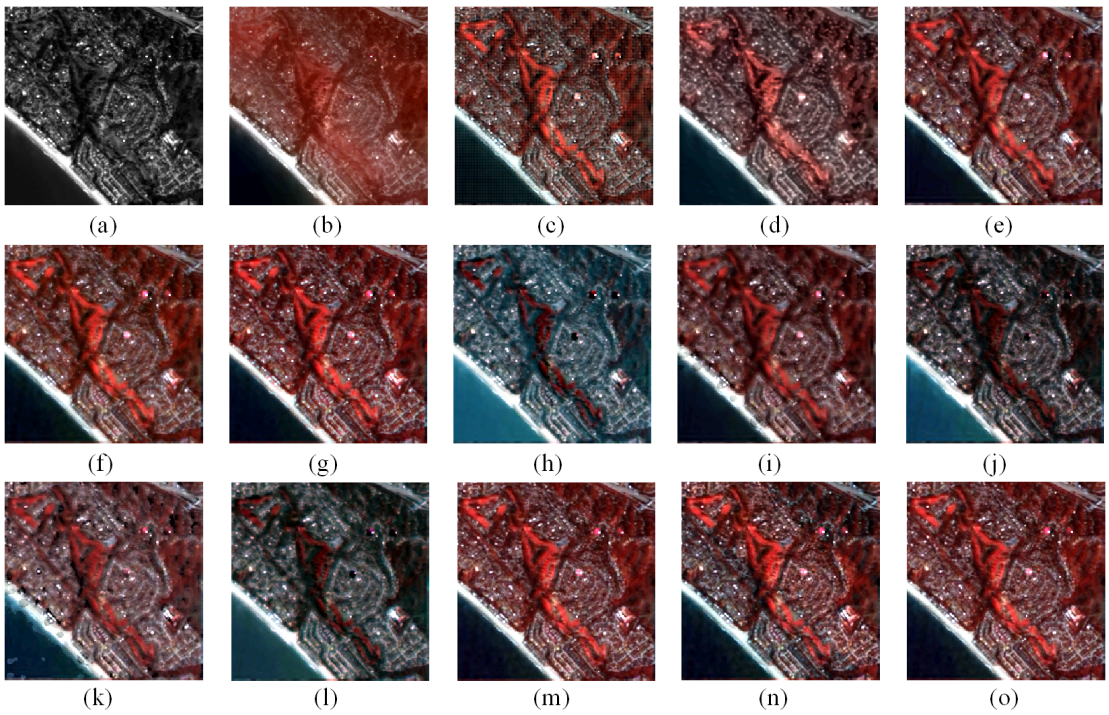


Fig. 6. Pansharpening results on Hyperion2 data. (a) Real PAN image. (b) GFPCA. (c) Bayesian Naïve. (d) Bayesian sparse. (e) AWLP. (f) STF. (g) SEGM. (h) GIHS_original. (i) GIHS_proposed. (j) Brovey_original. (k) Brovey_proposed. (l) GS_original. (m) GS_proposed. (n) GSA_original. (o) GSA_proposed.

sparse, STF, and SEGM algorithms. In general, the original CS-based pansharpening algorithms suffer from more severe spectral distortion due to the unmatched spectral range between HS and PAN images. The application of the detail extraction procedure proposed in this article greatly improves the performance of some CS-based pansharpening methods, as shown by the visual inspection. The original GIHS, Brovey, GS, and GSA provide relatively poor performance, both from subjective visual inspection and objective numerical points of view, when compared to their improved versions that use the same injection methodologies but different detail extraction procedures. It is worth noting that the performance of the improved CS-based pansharpening algorithms is, in general, also better than or comparable to several advanced algorithms for comparison, including Bayesian Naïve, Bayesian sparse, AWLP, STF, and SEGM algorithms. Overall, the improved versions of the CS-based methods show good spatial and spectral performance on the two real data sets when compared against the original CS-based and the other state-of-the-art pansharpening methods. Note that, the proposed detail extraction method plays an important role in providing good performance in terms of spectral fidelity and spatial sharpness.

The objective evaluation indexes of different algorithms on real data sets Hyperion1 and Hyperion2 at the reduced resolution are summarized in Table II. Consistent with subjective analysis, the objective performances of AWLP, STF, and SEGM algorithms are better than that of GIHS, Brovey, GS, and GSA algorithms. When we use the detail extraction procedure proposed in this article to replace the detail extraction steps of the original GIHS, Brovey, GS, and GSA algorithms, the performances of these algorithms are greatly improved. Almost all best and second-best values are obtained by improved versions of these algorithms. In other words, the proposed algorithm greatly improves the performance of the original GIHS, Brovey, GS, and GSA algorithms, and the performance of the improved versions of these four algorithms is superior to that of the other state-of-the-art methods, which further proves that the proposed method can offer a good tradeoff between spectral fidelity and spatial sharpness.

Table III summarizes objective evaluation indexes of different pansharpening algorithms at full resolution on real data sets Hyperion1 and Hyperion2. The objective analysis at full resolution is in line with that of the reduced resolution. The improved version of the GSA method provides all the largest QNR values on both Hyperion1 and Hyperion2 data sets. Again, the proposed CS-based pansharpening algorithms show improvement of spectral fidelity confirmed by the values of D_λ. The D_λ of the improved GSA ranks first on Hyperion1 data set and Hyperion2 data set. The objective quality evaluation of the improved GS is still slightly poorer than that of the improved GSA method. In general, the proposed algorithm achieves good objective performance at full resolution evaluation. The superiority of the detail extraction procedure evidenced in the subjective visual analysis is still confirmed at both the reduced resolution and full resolution.

The computational complexity of each method is also compared, and the fusion experiments are all implemented in MATLAB on a computer with Intel Core i5-7300HQ CPU at 2.50 GHz and 8-GB memory. The average running times of different algorithms are tabulated in Table IV, in which these times are the average values of the computing times of all data sets at the reduced resolution experiments. As shown

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TABLE III
QUANTITATIVE ASSESSMENT FOR THE EXPERIMENTAL SYNTHETIC DATA SETS

Data set		Hyperion1			Hyperion2		
Index		D_λ	D_S	QNR	D_λ	D_S	QNR
Reference		0	0	1	0	0	1
GFPCA		0.0399	0.0174	0.9434	0.0202	0.0146	0.9655
Bayesian Naïve		0.0165	0.0096	0.9741	0.0083	0.0092	0.9826
Bayesian Sparse		0.0213	0.0122	0.9667	0.0128	0.0115	0.9758
AWLP		0.0456	0.0116	0.9434	0.0213	0.0145	0.9645
STF		0.0342	0.0223	0.9442	0.0125	0.0226	0.9652
SEGM		0.0279	0.0124	0.9601	0.0086	0.0088	0.9827
GIHS	original	0.0285	0.0132	0.9587	0.0228	0.0116	0.9659
	proposed	0.0196	0.0095	0.9711	0.0158	0.0116	0.9727
Brovey	original	0.0539	0.0152	0.9310	0.0194	0.0116	0.9693
	proposed	0.0309	0.0095	0.9599	0.0129	0.0105	0.9767
GS	original	0.0227	0.0110	0.9665	0.0178	0.0169	0.9656
	proposed	0.0206	0.0106	0.9690	0.0085	0.0093	0.9823
GSA	original	0.0424	0.0099	0.9481	0.0108	**0.0081**	0.9812
	proposed	**0.0109**	**0.0089**	**0.9803**	**0.0066**	0.0084	**0.9851**

TABLE IV
AVERAGE COMPETING TIME (S) OF DIFFERENT ALGORITHMS

GFPCA	Bayesian Naïve	Bayesian Sparse	AWLP	STF	SEGM	GIHS	Brovey	GS	GSA	GIHS_pro	Brovey_pro	GS_pro	GSA_pro
2.63	1.87	37.75	1.35	6.08	2.11	0.41	0.46	1.73	1.97	19.59	19.14	20.48	20.94

in Table IV, the GFPCA, Bayesian Naïve, GIHS, Brovey, GS, and GSA methods are very fast, but the fused images obtained by these methods are unsatisfactory. The four improved CS-based algorithms by applying the proposed method are slower than the AWLP, SEGM, and STF algorithms, and are faster than the Bayesian sparse algorithm. The proposed CS-based algorithms are not comparable from the running time point of view, but the fusion performance of the proposed algorithm is comparable to them, especially for two sets of real data. More importantly, the improved versions by using the proposed method all provide much better fusion performance than the original CS-based algorithms. Most of the processes of the proposed method are simple and fast. For example, the fusion rules of the salient regions and the nonsalient regions are fast and have an $O(n)$ time. However, the GMM-based segmentation threshold selection method requires iterative optimization and is time-consuming. The proposed fusion algorithm needs adequate computation time to assign the fusion processes and has polynomial complexity, which is considered an acceptable algorithm. We will improve the GMM optimization iteration method, and develop parallel processing method to accelerate the running speed of the proposed algorithm in future studies.

F. Validity Discussion of the LOG Enhancement Algorithm on the PAN Image

This section analyzes the effect of the preprocessing, i.e., the LOG enhancement algorithm on the PAN image. Since the proposed method is applied to four CS-based methods, we test on the Moffett field data set to compare these four improved versions with two groups of experiments, where the input PAN image is, respectively, not enhanced and enhanced by using the LOG algorithm. Table V summarizes the experimental results of the quantitative analysis. As discussed in Section IV-C, the Moffett field data set is a synthetic data, and CC, SAM, ERGAS, and Q index are used to analyze the performance. From Table V, we can see that when the PAN image is enhanced by the LOG enhancement algorithm, the performance is better than that without the enhancement algorithm. Experimental results in Table V demonstrate that the LOG enhancement algorithm is an effective preprocessing step for HS pansharpening.

TABLE V
QUANTITATIVE ASSESSMENT OF THE LOG ENHANCEMENT STRATEGY

Index		CC	SAM	ERGAS	Q
GIHS_proposed	No-enhance	0.9601	**7.4360**	4.9255	0.8831
	LOG	**0.9602**	7.4471	**4.9217**	**0.8833**
Brovey_proposed	No-enhance	0.9598	**7.3223**	4.9864	**0.8819**
	LOG	**0.9600**	**7.3223**	**4.9781**	0.8817
GS_proposed	No-enhance	0.9534	**7.3471**	5.2327	0.8832
	LOG	**0.9602**	7.4269	**4.8979**	**0.8844**
GSA_proposed	No-enhance	0.9620	7.2746	4.9405	0.8850
	LOG	**0.9695**	**7.2432**	**4.5203**	**0.8866**

G. Validity Discussion of the GMM-Based Threshold Selection Method

In order to demonstrate the validity of the GMM-based threshold selection method, we compare the fusion performance of the proposed method using the thresholds obtained by different thresholding techniques, including the proposed GMM-based threshold selection method, the powerful Otsu

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TABLE VI
QUANTITATIVE ASSESSMENT OF THE THRESHOLD SELECTION METHODS

Index		CC	SAM	ERGAS	Q
GIHS	Mean	0.9537	**7.3439**	5.2810	0.8824
	Otsu	0.9539	7.3452	5.2709	0.8824
	proposed	**0.9602**	7.4471	**4.9217**	**0.8833**
Brovey	Mean	0.9537	**7.3223**	5.3242	0.8812
	Otsu	0.9539	7.3224	5.3184	0.8812
	proposed	**0.9600**	**7.3223**	**4.9781**	**0.8817**
GS	Mean	0.9532	**7.3658**	5.2819	0.8813
	Otsu	0.9534	7.3662	5.2722	0.8814
	proposed	**0.9602**	7.4269	**4.8979**	**0.8844**
GSA	Mean	0.9721	7.3735	4.5642	0.8852
	Otsu	**0.9722**	7.4326	4.5604	0.8857
	proposed	0.9695	**7.2432**	**4.5203**	**0.8866**

thresholding technique, and the basic statistical feature mean. Table VI shows the quantitative assessment of three different threshold selection methods on the Moffett field data set. It can be seen that the performance of the proposed pansharpening algorithm using a GMM-based threshold selection method is significantly better than that of the other two thresholding techniques. This is because the GMM divides an image into several clusters and utilizes multiple Gaussian distributions to characterize and approximate the image. The statistical parameters obtained by the GMM are more suitable for image description. Therefore, the GMM-based threshold selection algorithm can be used for processing the images which contain multiple different objects. This is not always possible for the basic statistical feature mean thresholding method which relies heavily on having the evenly distributed grayscale values. The Otsu method may have unstable performance when processing images with a darker background.

V. CONCLUSION

HS pansharpening essentially aims at improving the spatial resolution of the HS image. This article proposes a detail extraction algorithm for HS pansharpening. The difference between the proposed algorithm and the traditional CS-based algorithms is mainly reflected in two aspects: on the one hand, we fuse the details of PAN and that of HS images to obtain the total detail map and use this detail map to match and replace the spatial component of the HS image, which may be beneficial to reduce the spectral distortion caused by spectral range mismatch in traditional CS-based pansharpening methods; on the other hand, we divide the image into salient and nonsalient regions, and only utilize the salient region of the PAN image for the detail extraction, which reduces the spectral distortion caused by the low-pass component of the PAN image.

The proposed algorithm has been implemented into four credible pansharpening methods including the GIHS, Brovey, GS, and GSA methods. Compared with several state-of-the-art pansharpening algorithms, the experimental results show that the proposed method improves the spatial quality of the pansharpened image while effectively preserving the spectral fidelity of the HS image. The proposed algorithm is based on the principle of CS pansharpening methods to replace the spatial information of the original HS image with the extracted spatial information of the HS and PAN images. Therefore, the injection rules in the MRA pansharpening methods cannot be implemented into the proposed algorithm. In the future, the plan is to research whether the proposed algorithm can be improved further to be adapted to MRA pansharpening methods.

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